

HALF-SESSION TRADING: MAXIMIZING SHARPE WITH MACHINE LEARNING

OHLC & News-Driven Risk-Adjusted Trading

NAME OF PROJECT:
HalfLife Alpha

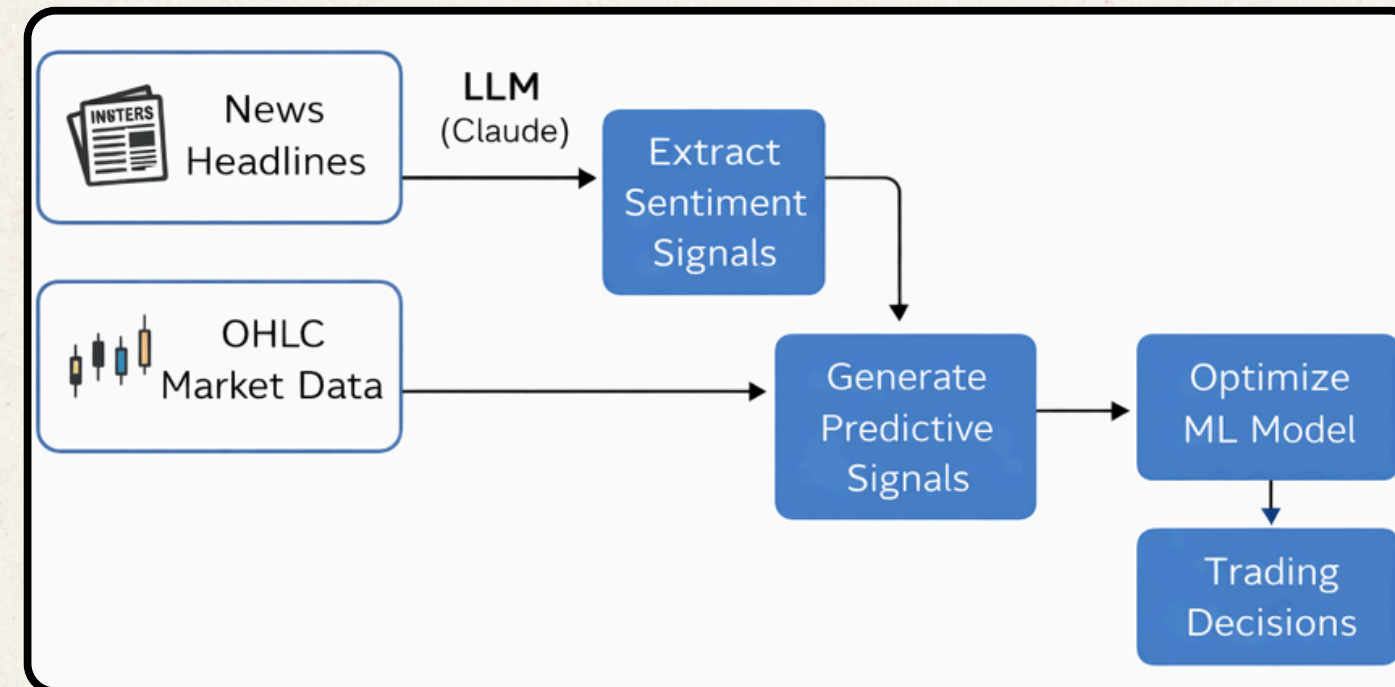
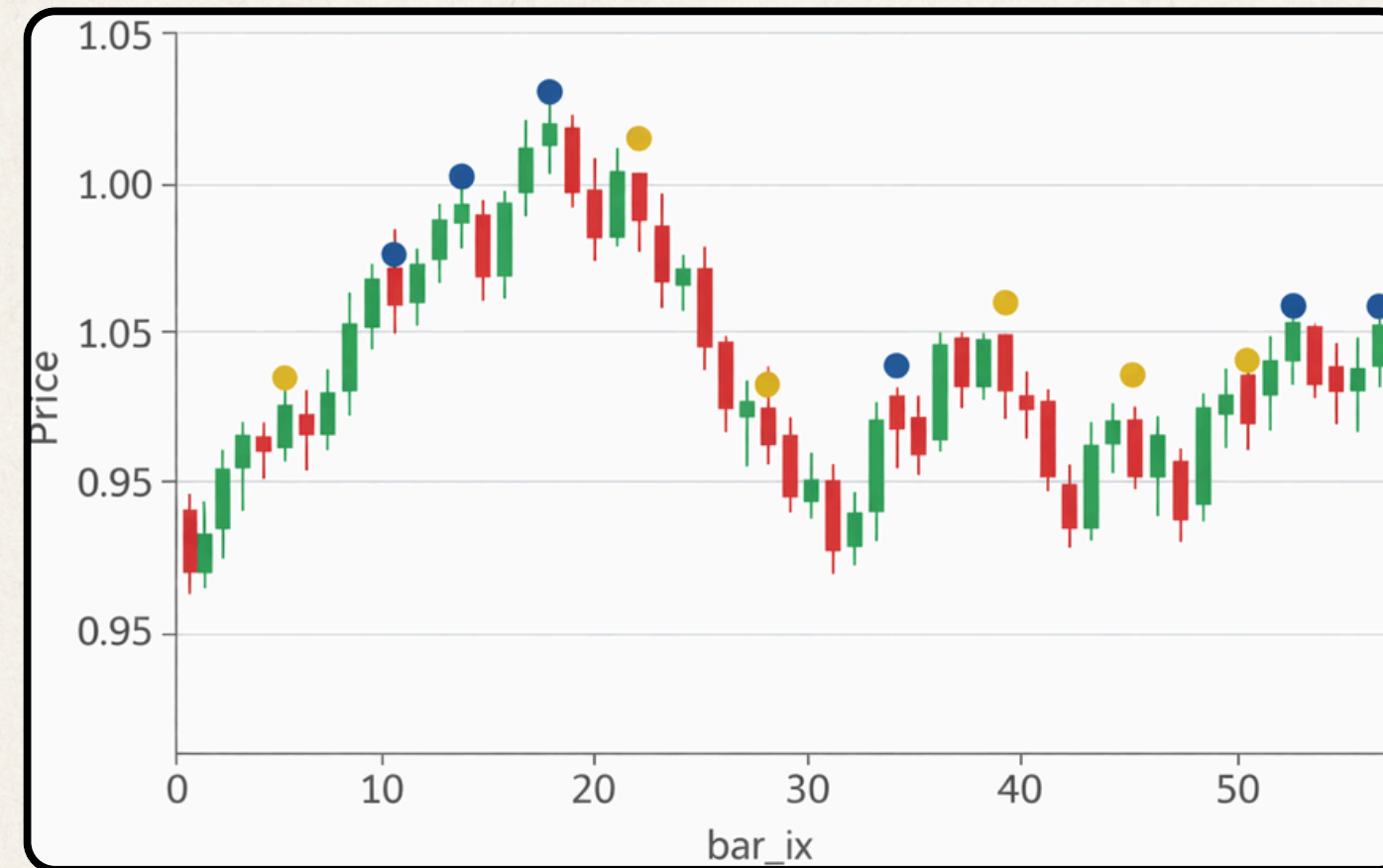
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Overview

We develop a **machine learning-based** trading strategy using partial market data (**OHLC and headlines**) to predict optimal positions.

The objective is to **maximize risk-adjusted returns**, measured by the Sharpe ratio.



- 01** We use **LLMs (Claude)** to extract sentiment from headlines, quantify their **correlation with OHLC price dynamics**, and transform these signals into **long-term** indicators for trading decisions.
- 02** We evaluate multiple baseline **ML models** on OHLC features and select the optimal model by **optimizing Sharpe ratio performance**.
- 03** We **integrate headlines-derived signals with OHLC features** and iteratively refine the model to optimize trading performance.